

DOTA Dataset with OBB

DOTA stands as a specialized dataset, emphasizing **object detection** in aerial images. Originating from the DOTA series of datasets, it offers annotated images capturing a diverse array of aerial scenes with Oriented Bounding Boxes (OBB).



Key Features

- Collection from various sensors and platforms, with image sizes ranging from 800×800 to $20,000 \times 20,000$ pixels.
- Features more than 1.7M Oriented Bounding Boxes across 18 categories.
- Encompasses multiscale object detection.
- Instances are annotated by experts using arbitrary (8 d.o.f.) quadrilateral, capturing objects of different scales, orientations, and shapes.

Dataset Versions

DOTA-v1.0

- Contains 15 common categories.
- Comprises 2,806 images with 188,282 instances.
- Split ratios: 1/2 for training, 1/6 for validation, and 1/3 for testing.

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DOTA-v1.5

- Incorporates the same images as DOTA-v1.0.
- Very small instances (less than 10 pixels) are also annotated.
- Addition of a new category: "container crane".
- A total of 403,318 instances.
- Released for the DOAI Challenge 2019 on Object Detection in Aerial Images.

DOTA-v2.0

- Collections from Google Earth, GF-2 Satellite, and other aerial images.
- Contains 18 common categories.
- Comprises 11,268 images with a whopping 1,793,658 instances.
- New categories introduced: "airport" and "helipad".
- Image splits:
 - Training: 1,830 images with 268,627 instances.
 - Validation: 593 images with 81,048 instances.
 - Test-dev: 2,792 images with 353,346 instances.
 - Test-challenge: 6,053 images with 1,090,637 instances.

Dataset Structure

DOTA exhibits a structured layout tailored for OBB object detection challenges:

- **Images:** A vast collection of high-resolution aerial images capturing diverse terrains and structures.
- **Oriented Bounding Boxes:** Annotations in the form of rotated rectangles encapsulating objects irrespective of their orientation, ideal for capturing objects like airplanes, ships, and buildings.

Applications

DOTA serves as a benchmark for training and evaluating models specifically tailored for aerial image analysis. With the inclusion of OBB annotations, it provides a unique challenge, enabling the development of specialized object detection models that cater to aerial imagery's nuances.

Dataset YAML



Typically, datasets incorporate a YAML (Yet Another Markup Language) file detailing the dataset's configuration. For DOTA v1 and DOTA v1.5, Ultralytics provides `DOTAv1.yaml` and `DOTAv1.5.yaml` files. For additional details on these as well as DOTA v2 please consult DOTA's official repository and documentation.



DOTAv1.yaml

```
# Train/val/test sets as 1) dir: path/to/imgs, 2) file: path/to/imgs.txt, or
# 3) list: [path/to/imgs1, path/to/imgs2, ..]
path: ../datasets/DOTAv1 # dataset root dir
train: images/train # train images (relative to 'path') 1411 images
val: images/val # val images (relative to 'path') 458 images
test: images/test # test images (optional) 937 images

# Classes for DOTA 1.0
names:
  0: plane
  1: ship
  2: storage tank
  3: baseball diamond
  4: tennis court
  5: basketball court
  6: ground track field
  7: harbor
  8: bridge
  9: large vehicle
  10: small vehicle
  11: helicopter
  12: roundabout
  13: soccer ball field
  14: swimming pool

# Download script/URL (optional)
```

Split DOTA images

To train DOTA dataset, we split original DOTA images with high-resolution into images with 1024x1024 resolution in multiscale way.

 Split images**Python**

```
from ultralytics.data.split_dota import split_test, split_trainval

# split train and val set, with labels.
split_trainval(
    data_root="path/to/DOTAv1.0/",
    save_dir="path/to/DOTAv1.0-split/",
    rates=[0.5, 1.0, 1.5], # multiscale
    gap=500,
)
# split test set, without labels.
split_test(
    data_root="path/to/DOTAv1.0/",
    save_dir="path/to/DOTAv1.0-split/",
    rates=[0.5, 1.0, 1.5], # multiscale
    gap=500,
)
```

Usage

To train a model on the DOTA v1 dataset, you can utilize the following code snippets. Always refer to your model's documentation for a thorough list of available arguments.

 Warning

Please note that all images and associated annotations in the DOTA v1 dataset can be used for academic purposes, but commercial use is prohibited. Your understanding and respect for the dataset creators' wishes are greatly appreciated!

Train Example

Python

```
from ultralytics import YOLO

# Create a new YOLO11n-OBb model from scratch
model = YOLO("yolo11n-obb.yaml")

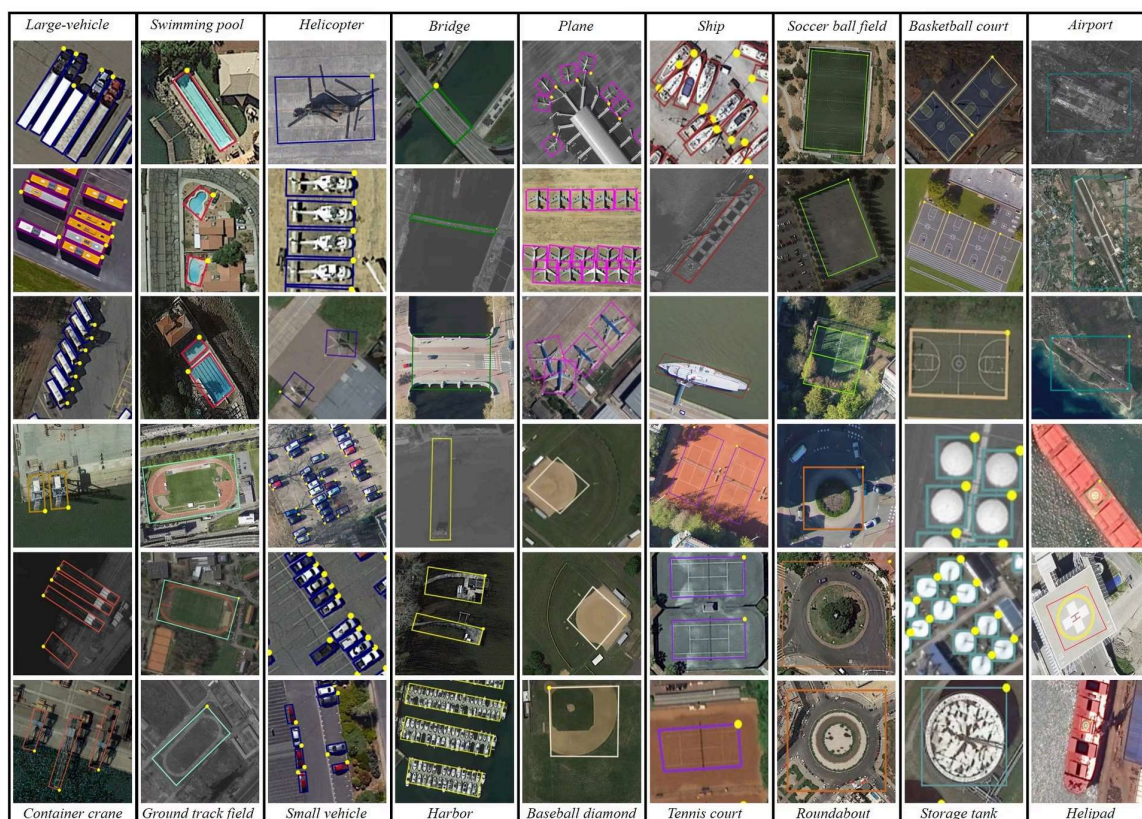
# Train the model on the DOTA v1 dataset
results = model.train(data="DOTAv1.yaml", epochs=100, imgsz=1024)
```

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```
# Train a new YOLO11n-OBb model on the DOTA v1 dataset
yolo obb train data=DOTAv1.yaml model=yolo11n-obb.pt epochs=100 imgsz=1024
```

Sample Data and Annotations

Having a glance at the dataset illustrates its depth:



- **DOTA examples:** This snapshot underlines the complexity of aerial scenes and the significance of Oriented **Bounding Box** annotations, capturing objects in their natural orientation.



The dataset's richness offers invaluable insights into object detection challenges exclusive to aerial imagery.

Citations and Acknowledgments

For those leveraging DOTA in their endeavors, it's pertinent to cite the relevant research papers:

BibTeX

```
@article{9560031,  
  author={Ding, Jian and Xue, Nan and Xia, Gui-Song and Bai, Xiang and Yang,  
Wen and Yang, Michael and Belongie, Serge and Luo, Jiebo and Datcu, Mihai and  
Pelillo, Marcello and Zhang, Liangpei},  
  journal={IEEE Transactions on Pattern Analysis and Machine Intelligence},  
  title={Object Detection in Aerial Images: A Large-Scale Benchmark and  
Challenges},  
  year={2021},  
  volume={},  
  number={},  
  pages={1-1},  
  doi={10.1109/TPAMI.2021.3117983}  
}
```

A special note of gratitude to the team behind the DOTA datasets for their commendable effort in curating this dataset. For an exhaustive understanding of the dataset and its nuances, please visit the [official DOTA website](#).

FAQ

What is the DOTA dataset and why is it important for object detection in aerial images?

The [DOTA dataset](#) is a specialized dataset focused on object detection in aerial images. It features Oriented Bounding Boxes (OBB), providing annotated images from diverse aerial scenes. DOTA's diversity in object orientation, scale, and shape across its 1.7M annotations and 18 categories makes it ideal for developing and evaluating models tailored for aerial imagery analysis, such as those used in surveillance, environmental monitoring, and disaster management.

How does the DOTA dataset handle different scales and orientations in images?

DOTA utilizes Oriented Bounding Boxes (OBB) for annotation, which are represented by rotated rectangles encapsulating objects regardless of their orientation. This method ensures  **EN**

objects, whether small or at different angles, are accurately captured. The dataset's multiscale images, ranging from 800×800 to $20,000 \times 20,000$ pixels, further allow for the detection of both small and large objects effectively.

How can I train a model using the DOTA dataset?

To train a model on the DOTA dataset, you can use the following example with Ultralytics YOLO:

Train Example

Python

```
from ultralytics import YOLO

# Create a new YOLO11n-OBb model from scratch
model = YOLO("yolo11n-obb.yaml")

# Train the model on the DOTA v1 dataset
results = model.train(data="DOTAv1.yaml", epochs=100, imgsz=1024)
```

CLI

```
# Train a new YOLO11n-OBb model on the DOTA v1 dataset
yolo obb train data=DOTAv1.yaml model=yolo11n-obb.pt epochs=100 imgsz=1024
```

For more details on how to split and preprocess the DOTA images, refer to the [split DOTA images section](#).

What are the differences between DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0?

- **DOTA-v1.0:** Includes 15 common categories across 2,806 images with 188,282 instances. The dataset is split into training, validation, and testing sets.
- **DOTA-v1.5:** Builds upon DOTA-v1.0 by annotating very small instances (less than 10 pixels) and adding a new category, "container crane," totaling 403,318 instances.
- **DOTA-v2.0:** Expands further with annotations from Google Earth and GF-2 Satellite, featuring 11,268 images and 1,793,658 instances. It includes new categories like "airport" and "helipad."

For a detailed comparison and additional specifics, check the [dataset versions section](#).

How can I prepare high-resolution DOTA images for training?

DOTA images, which can be very large, are split into smaller resolutions for management and training. Here's a Python snippet to split images:



 Example**Python**

```
from ultralytics.data.split_dota import split_test, split_trainval

# split train and val set, with labels.
split_trainval(
    data_root="path/to/DOTAv1.0/",
    save_dir="path/to/DOTAv1.0-split/",
    rates=[0.5, 1.0, 1.5], # multiscale
    gap=500,
)
# split test set, without labels.
split_test(
    data_root="path/to/DOTAv1.0/",
    save_dir="path/to/DOTAv1.0-split/",
    rates=[0.5, 1.0, 1.5], # multiscale
    gap=500,
)
```

This process facilitates better training efficiency and model performance. For detailed instructions, visit the [split DOTA images section](#).



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